

Acne and Sebaceous Glands

Acne loci GWAS, androgens, SHBG and 5-alpha-reductase, IGF-1 and mTORC1, follicle inflammation, hidradenitis, sebaceous and rosacea cysts, and isotretinoin

Sérgio F · 03-10-2026 · local time

Informative and educational genetic report, based only on Sérgio F genes (genotyping chip + local imputation). It is neither a diagnosis nor a prescription. All actions are marked as suggestion (not applied).

Contents

Section	Page
1. Summary	3
2. How to read this report	4
3. Polygenic scores	5
4. Key variants by topic	6
5. GWAS loci burden	9
6. Favourable	10
7. Unfavourable	11
8. Mechanisms of ageing	12
9. Practical suggestions	13
10. Limitations	14
11. Methods and sources	15
12. Annex: all variants analysed	16

1. Summary

Favourable / typical. Acne has a typical genetic predisposition: the 91 loci GWAS load is at percentile 35.4 and the polygenic score, developed in another population, in the 16.0. The hormonal axis is favourable: SHBG high genetics (percentile 82.6, with 1 allele of SHBG higher in rs6259 and rs1799941), which reduces the free testosterone that reaches the sebaceous gland. Genetic testosterone is typical (58.0) and the genetic IGF-1 is slightly low (32.1). Sebaceous cysts (17.5) and rosacea (11.7) have low loci loads. CYP2C8 is normal: isotretinoin is metabolized in a typical way.

Unfavourable. It has homozygosity for the risk allele in three strong follicle loci (**SEMA4B**, **WNT10A** and **FGF10**) and 1 allele in BMAL1, FST, LYPLAL1 and CLEC16A. These are already counted in the typical load. The suppurative hidradenitis has a loci load at percentile 92.9, with 1 allele in the SOX9 locus. IL1A -889 and FTO are heterozygous.

What does it mean? Weekly isotretinoin already acts at the strongest point of acne genetics, sebaceous gland and follicle. Genes don't ask for any more acne. The point of attention is the high load of hidradenitis: to know how to recognize the first nodules in the armpits and in the groins.

2. How to read this report

- **Data:** Sérgio F chip genotypes (GRCh38, strand +) and, when SNP is not in the chip, locally imputed genotypes (Beagle 5.5, public panel of genomes complete to 30x) with quality $DR2 \geq 0.8$. Under this threshold SNP is treated as SNP not found in the data.
- **Key variants** 21 Selected SNPs from the literature (GWAS Catalog, ClinVar, dbSNP and reference studies): 12 of the chip, 9 imputed and 0 not found.
- **Strand:** All genotypes are in the GRCh38 strand +. The published effect allele is compared with the observed alleles; 0 SNPs needed conversion to the complementary strand and 4 are palindromic (A/T or C/G), solved by the reference allele and the position. When the literature uses the strand of the gene, the letters may differ from those of this report.
- **Polygenic scores (PGS):** 4 PGS Catalog scores calculated locally with 404,472 variants in total (chip + imputed). The percentile compares Sérgio F with the reference population ($n = 633$) calculated with exactly the same variants. Percentage ≥ 80 : MORE LIKELY/HIGH; ≤ 20 : LESS LIKELY/LOW; between 20 and 80: TYPICAL.
- **Loci load GWAS:** For each trait, all SNPs with $p \leq 5 \times 10^{-8}$ in GWAS Catalog (1 line for SNP, the most significant) are added as simple count of effect alleles (each allele valley 1). The sum of Sérgio F is compared with that of the reference population ($n = 633$) with exactly the same variants, which gives an empirical percentile. Many of these SNPs are in linkage disequilibrium, so the load is a complement, not a substitute, of the polygenic scores. 167 loci, 154 with genotype.
- **Judgment:** Each result is divided into Favourable or Unfavourable (or typical/without judgment). A genotype is a probability, not a destination; almost all effects are small and add to the lifestyle.
- **Evidence:** Strong : replicated and with clear clinical effect; Moderate = GWAS/consistent meta-analysis, modest effect; Weakness : small or mechanistic studies.

3. Polygenic scores

In the acne score, a higher percentile = more risk. In hormonal scores, the percentile only says if the level tends to be naturally higher or lower; it is not risk by itself.

Acne

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Acne	PGS019635 (Chen 2025, LDpred2, 274 thousand variants)	267,135 273,955 (weighted coverage 97%)	16.0	LESS LIKELY	favourable

- **Acne:** Score developed in another population: reduced transferability; reading only as an indication.

Hormones that feed the tallow

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Testosterone (men)	PGS001988 (Privé 2022, 3985 variants)	3,857 3,985 (weighted coverage 97%)	58.0	TYPICAL	without judgment
SHBG	PGS001977 (Privé 2022, 51.4 thousand variants)	49,749 51,427 (weighted coverage 97%)	82.6	HIGH	without judgment
IGF-1	PGS001960 (Privé 2022, 87.1 thousand variants)	83,731 87,149 (weighted coverage 96%)	32.1	TYPICAL	without judgment

- **SHBG:** High SHBG = less testosterone free for the sebaceous gland.
- **IGF-1:** IGF-1 activates the mTORC1 and the production of sebum.

4. Key variants by topic

Variants with described effect. The judgment refers to acne and follicle diseases; in hormones, high SHBG is favourable for the skin.

Acne loci GWAS (pilous follicle and sebaceous gland)

Loci of the largest GWAS meta-analysis of acne. Many genes are linked to the development of hair follicle and sebaceous gland (WNT10A, FGF10, FST, SEMA4B), more than hormones. This explains why acne has a high inheritance (~80% in twins).

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
SEMA4B	rs34560261	C/C	chip	chip	C (2 copy/s)	unfavourable	The strongest locus of acne (semaphorin 4B, follicle) [GWAS Catalog: acne vulgaris, allele C, OR 1.25, p=3E-35, PMID 35132056]	Moderate
BMAL1/ RASSF10	rs3849154	T/G	imputation	DR2=0.99	T (1 copy/s)	unfavourable (light)	11p15 (near BMAL1, circadian clock gene) [GWAS Catalog: acne vulgaris, allele T, OR 1.15, p=2E-28, PMID 35132056]	Moderate
FST	rs629725	C/T	imputation	DR2=0.98	T (1 copy/s)	unfavourable (light)	Follistatin (differentiation of hair follicle) [GWAS Catalog: acne vulgaris, allele T, OR 1.15, p=4E-28, PMID 35132056]	Moderate
LYPLAL1	rs1256580	G/C	imputation	DR2=0.99	C (1 copy/s)	unfavourable (light)	LYPLAL1 (lipids and fat distribution) [GWAS Catalog: acne vulgaris, allele C, OR 1.19, p=1E-27, PMID 35132056]	Moderate
WNT10A	rs74333950	T/T	chip	chip	T (2 copy/s)	unfavourable	WNT10A (follicle and gland development) [GWAS Catalog: acne vulgaris, allele T, OR 1.20, p=3E-23, PMID 35132056]	Moderate
FGF10	rs16874036	G/G	imputation	DR2=1.00	G (2 copy/s)	unfavourable	FGF10 (sebocytic FGFR2 signaling, the same pathway as Acne of Apert syndrome) [GWAS Catalog: acne vulgaris, allele G, OR 1.11, p=1E-15, PMID 35132056]	Moderate
CLEC16A	rs7194305	A/G	imputation	DR2=1.00	A (1 copy/s)	unfavourable (light)	CLEC16A (immunity and autophagy) [GWAS Catalog: acne vulgaris, allele A, OR 1.10, p=2E-15, PMID 35132056]	Weak

Androgens, SHBG and 5-alpha-reductase

Testosterone is converted to DHT by 5-alpha-reductase. DHT stimulates the sebaceous gland through the androgen receptor. SHBG holds testosterone in the blood: the more SHBG, the less free testosterone reaches the gland.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
SRD5A2	rs523349	V89L	C/C	chip	G (0 copy/s)	neutral	V89L (G = Leu89): less 5-alpha-reductase activity type 2 and less DHT	Weak
CYP17A1	rs743572	A1/A2	A/G	chip	G (1 copy/s)	unfavourable (light)	-34T>C promoter (A2 allele): associated in some studies with more androgens and acne	Weak
SHBG	rs6259	D356N	G/A	imputation	A (1 copy/s)	favourable	D356N (A = Asn): Higher SHBG, so less free testosterone	Moderate

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
SHBG	rs1799941		G/A	chip	A (1 copy/s)	favourable	Promoter: Allele A gives higher SHBG	Moderate
AR (X)	rs6152		G (X)	chip	A (0 copy/s)	without judgment	Androgen receptor (Stul, E211E): G allele is associated with more sensitivity to androgens in the follicle	Weak

Follicular inflammation

IL-1 α begins the formation of comedone. TNF- α and IGF-1/mTORC1 axis (where FTO also acts) increase inflammation and production of tallow.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
TNF	rs1800629	-308	G/G	chip	A (0 copy/s)	favourable	-308G>A: more TNF- α ; inconsistent association with acne	Weak
IL1A	rs1800587	-889	G/A	chip	A (1 copy/s)	unfavourable (light)	-889C>T (T of the gene = A in strand +): plus IL-1 α , which initiates comedogenesis	Weak
FTO	rs9939609		T/A	chip	A (1 copy/s)	unfavourable (light)	FTO: associated with more severe acne in some studies, via mTORC1 and metabolism	Weak

Suppurative and rosaceal hidradenitis

Suppurative hidradenitis is an inflammatory disease of the follicle in the folds (axillae, groins). Rosacea is a vascular and immune inflammation of the face, distinct from acne.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
SOX9 (17q24)	rs17226067		G/A	imputation DR2=0.98	G (1 copy/s)	unfavourable (light)	Stronger locus of suppurative hidradenitis (near SOX9, follicle) [GWAS Catalog: Suppurative hidradenitis, G allele, OR 1.28, p=5E-13, PMID 40650879]	Moderate
IRF1	rs6866614		G/G	chip	A (0 copy/s)	favourable	IRF1: rosacea locus [GWAS Catalog: Rosacea (PheCode 695.3), allele A, beta 0.123 [0.096-0.149] unit increase, p=4E-24, PMID 39024449]	Moderate
HERC2	rs12898729		G/G	imputation DR2=0.82	A (0 copy/s)	favourable	HERC2 (pigmentation; light skin) [GWAS Catalog: Rosacea (PheCode 695.3), allele G, beta 0.121 [0.089-0.153] unit decrease, p=3E-14, PMID 39024449] (the reported G allele reduces the trait; here the effect allele is the opposite, A)	Weak

Pharmacogenetics of isotretinoin

Isotretinoin is mainly metabolized by CYP2C8. Variants of loss of function would increase exposure and side effects.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
CYP2C8	rs11572080	*3 (R139K)	C/C	imputation DR2=1.00	T (0 copy/s)	favourable	CYP2C8*3 (R139K): isotretinoin is mainly metabolized by CYP2C8	Moderate
CYP2C8	rs10509681	*3 (K399R)	T/T	chip	C (0 copy/s)	favourable	CYP2C8*3 (K399R)	Moderate

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
CYP2C8	rs1058930	*4	G/G	chip	C (0 copy/s)	favourable	CYP2C8*4 (I264M)	Moderate

5. GWAS loci burden

Sum of all loci of GWAS Catalog ($p \leq 5 \times 10^{-8}$) of each disease, with each allele oriented towards the risk, compared to the reference population.

Trait (GWAS Catalog)	Loci (with genotype)	Sérgio F effect alleles	Calibrated variants	Percentile	Judgement
Acne	91 (88)	76 of 176	88	35.4	typical
Suppurative hidradenitis	25 (18)	18 of 36	18	92.9	unfavourable
Cysts and diseases of the sebaceous glands	42 (40)	28 of 80	40	17.5	favourable
Rosacea	9 (8)	5 of 16	8	11.7	favourable

6. Favourable

- **Acne with typical predisposition:** load of loci at percentile 35.4 (91 loci); score in the 16.0 (developed in another population, only indicative). **Moderate.**
- **High genetic SHBG** (percentile 82.6; SHBG D356N G/A and promoter G/A with 1 allele of SHBG higher): less testosterone free for the sebaceous gland. **Moderate.**
- **A slightly low genetic IGF-1** (percentile 32.1) and typical testosterone (58.0). **Weak.**
- **Cysts and diseases of the sebaceous glands** with loci at 17.5; rosacea in 11.7 (IRF1 and HERC2 without risk). **Moderate.**
- **TNF -308 G/G** (without the allele of more TNF- α). **Weak.**
- **CYP2C8 without *3 or *4:** typical exposure to isotretinoin. **Moderate.**

7. Unfavourable

- **Follicular loci in risk homozygosity:** SEMA4B C/C (OR ~1.25 by allele), WNT10A T/T and FGF10 G/G; 1 allele in BMAL1, FST, LYPLAL1 and CLEC16A. Already counted in the typical load. **Moderate.**
- **Suppurative hidradenitis:** load of loci at percentile 92.9 (18 loci used in calibration); SOX9 G/A with 1 allele. Weak to moderate (less loci).
- **IL1A -889 G/A, CYP17A1 A/G and FTO T/A** heterozygotes: small and inconsistent effects. **Weak.**
- **No judgment:** SRD5A2 V89 C/C (typical activity of 5-alpha-reductase); AR rs6152 allele G (X; more sensitivity to androgens in the follicle).

8. Mechanisms of ageing

Mapping in 84 mechanisms of aging of Sérgio F immortality objectives (only the mechanisms affected by this report).

Mechanism	Genetic finding	Meaning
Endocrine dysregulation	SHBG genetics at percentile 82.6; typical testosterone; Typical SRD5A2	Favourable
Nutrient Sensing Deregulation	IGF-1/mTORC1 of the sebocyte: IGF-1 genetic at percentile 32.1; FTO heterozygous	Neutral
Exhaustion of Stem Cells	Stem cells of the follicle and sebaceous gland: WNT10A, FGF10 and SEMA4B in risk homozygosity	Unfavourable (mild)
Chronic Inflammation	Hidradenitis with loci load at percentile 92.9; IL1A heterozygous; TNF -308 G/G	Unfavourable (mild)
Dysbiosis	Follicle and Cutibacterium acnes: no direct genetic marker	Neutral

9. Practical suggestions

All suggestions are suggestion (not applied), consistent with the current genetic plan, and should be discussed with the doctor.

1. **Weekly isotretinoin:** The normal CYP2C8 does not ask for adjustment because of the genes. Keep the lowest effective dose, as in the plan, with daily photoprotection and without retinol. *Suggestion (not applied).*
2. **Recognizing early hidradenitis:** with the loci load in the ~the 93rd percentile, painful and recurrent nodules in the axillae, in the groin or in the buttocks deserve early dermatological observation. Isotretinoin helps little in hidradenitis; there are specific treatments. Not smoking and broad clothes in the folds reduce outbreaks. *Suggestion (not applied).*
3. **Rapamycin:** mTOR inhibitors cause acneiform eruptions and folliculitis, especially in daily use. Register if 1-2 bubbles appear weeks after each dose; this is different from the genetic acne. *Suggestion (not applied).*
4. **Vitamins B:** High doses of B6 and B12 are described as the cause of acneiform eruptions. The plan already predicts to decrease the B6 to ≤ 5 mg; B12 of 1000 μ g is maintained. *Suggestion (not applied).*
5. **Diet (already in the plan):** Low glycemic hydrates and free sugars < 5% of energy lower insulin and IGF-1, which feed the sebocyte mTORC1. If bubbling with whey protein appears, try another source of protein. High fiber (the more the better). *Suggestion (not applied).*
6. **If it is considered finasteride or dutasteride for hair** (medical decision): reduce DHT and can decrease tallow. With high genetic SHBG, the androgen axis is already favourable for the skin. *Suggestion (not applied).*
7. **Face:** Non-comedogenic products; in seborrheic dermatitis, prefer moisturizers with squalane instead of vegetable oils. *Suggestion (not applied).*

Not recommended: testosterone, DHEA or secretagogues of GH/IGF-1 without medical indication (increased sebum and acne; the plan already avoids them); oral retinol or cod liver oil with isotretinoin; oral tetracyclines together with isotretinoin (intracranial hypertension).

10. Limitations

- There are no polygenic acne scores developed in the reference population. The score used comes from another population and is only indicative; the loci load is the main reading.
- The number of CAG repeats of the androgen receptor (which touches the sensitivity to androgens) is not read in a chip.
- The suppurative hidradenitis has few known loci (18 used in calibration), so the percentile of the load is unstable.
- Acne and sebum depend heavily on age, hormones, drugs (isotretinoin, rapamycin) and applied products.
- The load of loci GWAS sums SNPs in linkage imbalance. It is a descriptive complement and does not replace the polygenic scores.
- Conclusions only from genes; it is not a diagnosis.

11. Methods and sources

- **Genotypes:** Sérgio F chip (GRCh38, strand +). Absent SNPs come from local imputation (Beagle 5.5, public panel of complete 30x genomes), used only with $DR2 \geq 0.8$. No genetic data came out of the analysis computer.
- **Alleles and strand:** The alleles of each location are read from the Imputed VCF. The published effect allele is compared with those alleles and, if necessary, converted to the complementary strand (only in non-palindromic SNPs). In palindromic SNPs, strand + (GWAS Catalog standard and the harmonized files of PGS Catalog) is assumed. In the loci of GWAS Catalog with protective effect, the allele of effect is exchanged by the risk allele. In the X chromosome (AR), Sérgio F is hemizygous and has only one copy.
- **Polygenic scores:** Harmonized GRCh38 files from PGS Catalog: PGS019635 (Chen 2025), PGS001988, PGS001977 and PGS001960 (Privé 2022). Score = Δ coefficient $\times$ dose of effect allele. The percentile is empirical in the reference population ($n = 633$), with the same variants.
- **Loci load:** GWAS Catalog (associations with $p \leq 5 \times 10^{-8}$), alleles guided by the published effect signal, and empirical calibration in the same reference population.
- **Interpretation:** GWAS Catalog, ClinVar, dbSNP/Ensembl and reference literature: Mitchell 2022 and Teder-Laving 2023 (GWAS of acne), Navarini 2014, GWAS of suppurative hidradenitis and rosacea, reviews on mTORC1, IGF-1 and acne, and the label of isotretinoin. The complete references are available on request.

12. Annex: all variants analysed

Topic	Gene	SNP	Genotype	Source	Effect allele	Copies	Chain	Interpretation
Acne loci GWAS (pilous follicle and sebaceous gland)	SEMA4B	rs34560261	C/C	chip	C	2	strand +	2 risk allele copies
Acne loci GWAS (pilous follicle and sebaceous gland)	BMAL1/ RASSF10	rs3849154	T/G	imputation DR2=0.99	T	1	strand +	1 copy of risk allele
Acne loci GWAS (pilous follicle and sebaceous gland)	FST	rs629725	C/T	imputation DR2=0.98	T	1	strand +	1 copy of risk allele
Acne loci GWAS (pilous follicle and sebaceous gland)	LYPLAL1	rs1256580	G/C	imputation DR2=0.99	C	1	strand + (palindromic; assumed +)	1 copy of risk allele
Acne loci GWAS (pilous follicle and sebaceous gland)	WNT10A	rs74333950	T/T	chip	T	2	strand +	2 risk allele copies
Acne loci GWAS (pilous follicle and sebaceous gland)	FGF10	rs16874036	G/G	imputation DR2=1.00	G	2	strand +	2 risk allele copies
Acne loci GWAS (pilous follicle and sebaceous gland)	CLEC16A	rs7194305	A/G	imputation DR2=1.00	A	1	strand +	1 copy of risk allele
Androgens, SHBG and 5-alpha-reductase	SRD5A2	rs523349	C/C	chip	G	0	strand + (palindromic; assumed +)	without the protective allele (common genotype)
Androgens, SHBG and 5-alpha-reductase	CYP17A1	rs743572	A/G	chip	G	1	strand +	1 copy of risk allele
Androgens, SHBG and 5-alpha-reductase	SHBG	rs6259	G/A	imputation DR2=0.98	A	1	strand +	1 copy of the protective allele
Androgens, SHBG and 5-alpha-reductase	SHBG	rs1799941	G/A	chip	A	1	strand +	1 copy of the protective allele
Androgens, SHBG and 5-alpha-reductase	AR (X)	rs6152	G (X)	chip	A	0	strand +; hemizygous (chromosome X)	Allele A (X, hemizygous)
Follicular inflammation	TNF	rs1800629	G/G	chip	A	0	strand +	without the risk allele
Follicular inflammation	IL1A	rs1800587	G/A	chip	A	1	strand +	1 copy of risk allele
Follicular inflammation	FTO	rs9939609	T/A	chip	A	1	strand + (palindromic; assumed +)	1 copy of risk allele
Suppurative and rosacea hidradenitis	SOX9 (17q24)	rs17226067	G/A	imputation DR2=0.98	G	1	strand +	1 copy of risk allele
Suppurative and rosacea hidradenitis	IRF1	rs6866614	G/G	chip	A	0	strand +	without the risk allele
Suppurative and rosacea hidradenitis	HERC2	rs12898729	G/G	imputation DR2=0.82	A	0	strand +	without the risk allele
Pharmacogenetics of isotretinoin	CYP2C8	rs11572080	C/C	imputation DR2=1.00	T	0	strand +	without the risk allele
Pharmacogenetics of isotretinoin	CYP2C8	rs10509681	T/T	chip	C	0	strand +	without the risk allele
Pharmacogenetics of isotretinoin	CYP2C8	rs1058930	G/G	chip	C	0	strand + (palindromic; assumed +)	without the risk allele